

Fast Charger Slide Controller

Variant: [No Variations]

22/10/2024

Version P1

RELEASED 20-APR-2021

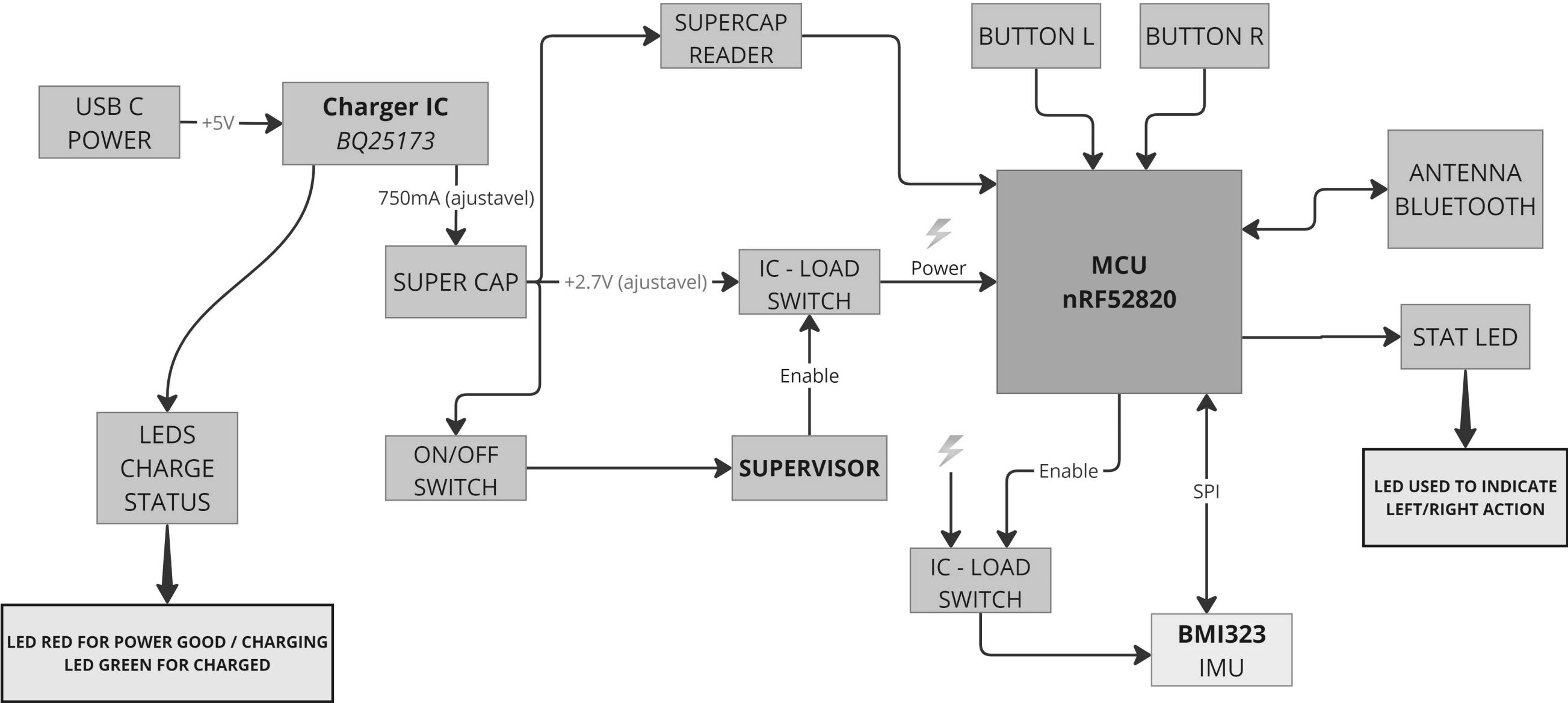
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TURY	EDGE		http://www.google.com	
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Fast Charger Slide Controller P1

(Block Diagram)



SUPERVIDOR is used to not allow use supercap when it reach some low voltage

IC SWITCH is used turn on/off the MCU. It's a low power IC.

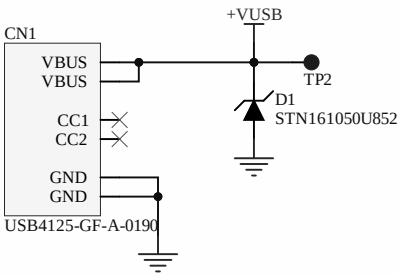
The MCU turns on the IMU when some of the buttons is kept pressed

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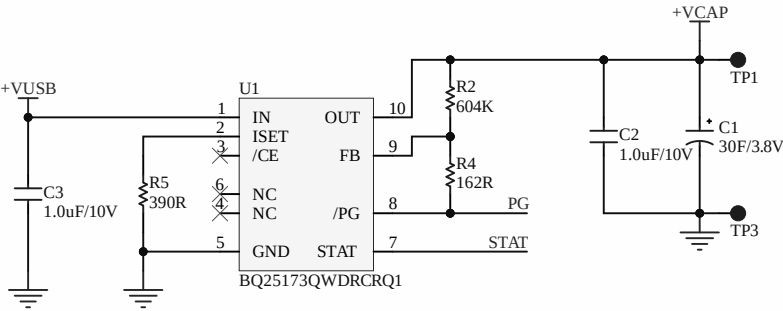
POWER SUPPLY



USB



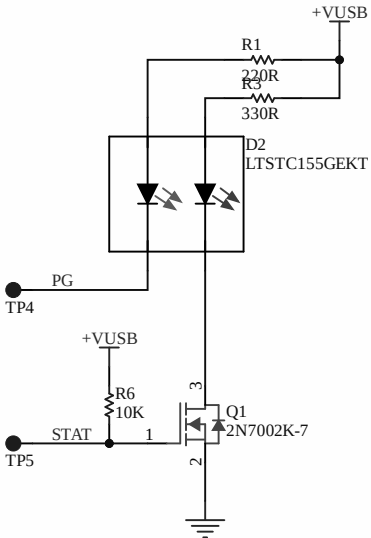
SUPERCAP CHARGER



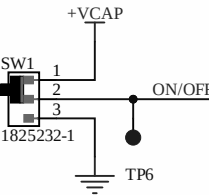
DESIGN NOTE:
/CE is used to enable the IC. Active LOW.
The IC has a intern pull down, so left
floating to always set enable.

DESIGN NOTE:
Resistor 390R in ISET set fast charge to
769mA.
The max allowed is 375R to set 800mA

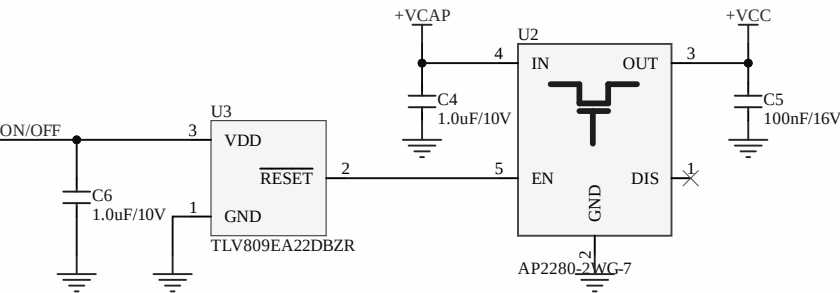
CHARGE STATUS LEDS



SWITCH ON/OFF



POWER LOAD SWITCH WITH VOLTAGE SUPERVISOR

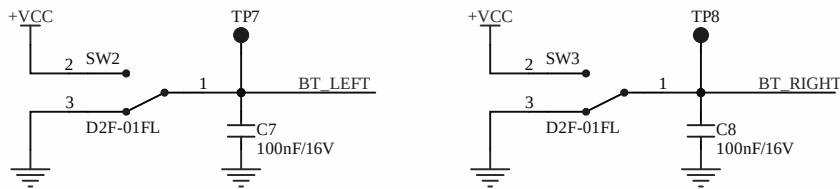


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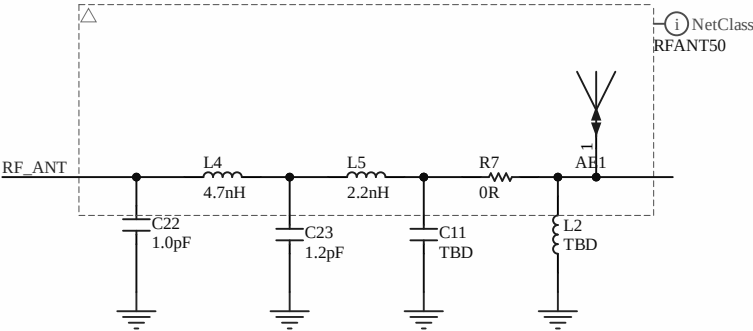
MICROCONTROLLER SCHEMATIC



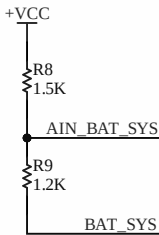
INPUT BUTTONS



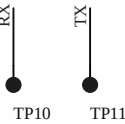
ANTENNA BLE



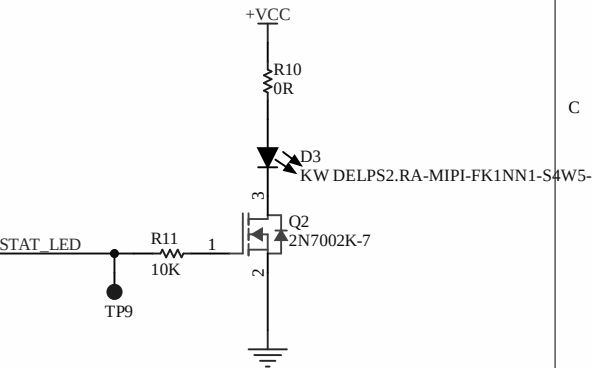
SUPERCAP READER



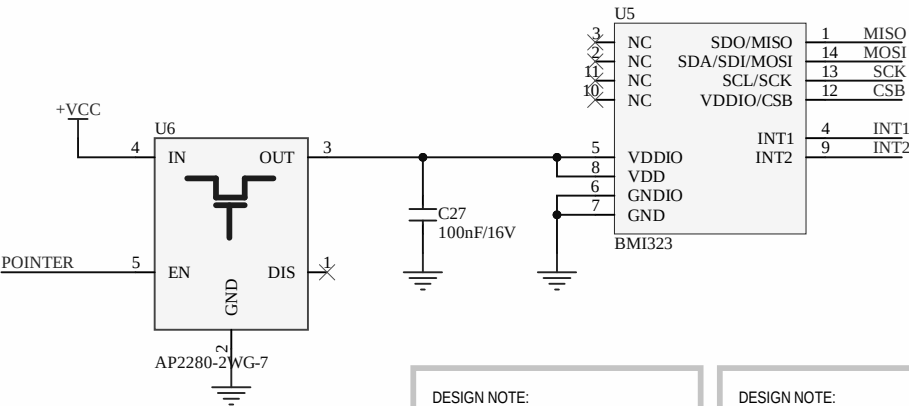
UART



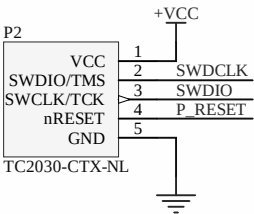
FEEDBACK LED



IMU - ACCEL/GYRO



SWD CONNECTOR



DESIGN NOTE:
PIN 1 receives energy from de SWD device to power on the MCU.

DESIGN NOTE:

The IMU is activated while some of the BUTTONs is kept pressed

DESIGN NOTE:

Pins INT1 e INT2 are used for interrupt mechanism, when SPI is set

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REVISION HISTORY								CENTRO DE INOVAÇÃO edge
SCHEMATIC PCB								
- Nullam pellentesque porttitor ex et dictum. Etiam vitae arcu eget nibh feugiat maximus. Curabitur ultrices lacinia tortor, vel gravida mauris viverra vel. - Duis posuere mi metus, ut imperdiet mauris eleifend id. Donec fermentum ut velit eu pellentesque. - Donec tincidunt augue eu massa scelerisque, nec auctor nisl faucibus. Praesent nec ex finibus, consequat velit sit amet, fermentum mauris. - Aliquam efficitur ante sit amet lobortis posuere. - Fusce luctus sapien sed felis porta rhoncus. Quisque a libero nec erat ultricies dictum. Sed leo eros, sollicitudin ut aliquet sit amet, imperdiet vitae metus. - Nunc vestibulum, risus quis dictum feugiat, neque odio gravida nulla, vel venenatis mi arcu sit amet sem. Mauris pulvinar aliquam nunc, nec hendrerit risus posuere vitae. - Nullam pellentesque porttitor ex et dictum. Etiam vitae arcu eget nibh feugiat maximus. Curabitur ultrices lacinia tortor, vel gravida mauris viverra vel. - Duis posuere mi metus, ut imperdiet mauris eleifend id. Donec fermentum ut velit eu pellentesque. - Donec tincidunt augue eu massa scelerisque, nec auctor nisl faucibus. Praesent nec ex finibus, consequat velit sit amet, fermentum mauris. - Aliquam efficitur ante sit amet lobortis posuere. - Fusce luctus sapien sed felis porta rhoncus. Quisque a libero nec erat ultricies dictum. Sed leo eros, sollicitudin ut aliquet sit amet, imperdiet vitae metus. - Nunc vestibulum, risus quis dictum feugiat, neque odio gravida nulla, vel venenatis mi arcu sit amet sem. Mauris pulvinar aliquam nunc, nec hendrerit risus posuere vitae.								
TURY EDGE Title: Fast Charger Slide Controller variants Page Contents: Revision: P1 Release Date: [No Variations] [05] - REVISION HISTORY.SchDoc Designed by: Rangel Gonçalves Checked by: Thiago L. 22/10/2024 Sheet: 5 of 6								
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Fast Charger Slide Controller - SCHEMATIC



Designator
[01] - COVER PAGE.SchDoc



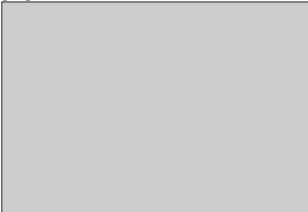
Designator
[02] - BLOCK DIAGRAM.SchDoc



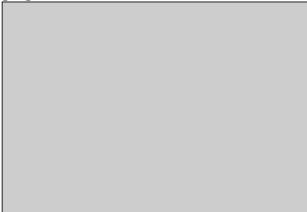
Designator
[03] - POWER SUPPLY.SchDoc



Designator
[04] - MICROCONTROLLER.SchDoc



Designator
[05] - REVISION HISTORY.SchDoc



FIDUCIALS TOOLING HOLES



	EDGE		http://www.google.com	
	TitlFast Charger Slide Controller			
Variants: [No Variations]				
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